

TEST OF MATHEMATICS FOR UNIVERSITY ADMISSION

PAPER 1: APPLICATIONS OF MATHEMATICAL KNOWLEDGE

Practice paper · Set 15

75 minutes

INSTRUCTIONS TO CANDIDATES

Please read these instructions carefully, but do not open the question paper until you are told that you may do so.

A separate answer sheet is provided for this paper. You require a soft pencil and an eraser. Please complete the answer sheet with your candidate number, centre number, date of birth and full name.

This paper contains **20 questions**. For each question, choose the **one** answer you consider correct and record your choice on the separate answer sheet. If you make a mistake, erase thoroughly and try again.

There are no penalties for incorrect responses, only marks for correct answers, so you should attempt all 20 questions. Each question is worth one mark.

You can use the question paper for rough working or notes, but no extra paper is allowed.

You must complete the answer sheet within the time limit.

Calculators and dictionaries are NOT permitted.

There is no formulae booklet for this test.

Please wait to be told you may begin before turning this page.

This is a practice paper generated from a question bank written to the TMUA content specification. It is not an official UAT-UK paper. Answers and worked solutions are in the accompanying solutions booklet.

BLANK PAGE

1 Given that $\sin \theta = \frac{3}{5}$ and θ is obtuse, find $\tan \theta$.

A $-\frac{3}{4}$

B $\frac{4}{3}$

C $-\frac{4}{3}$

D $-\frac{3}{5}$

E $\frac{3}{4}$

F $-\frac{4}{5}$

- 2 A length is given as 4.6 cm correct to 1 decimal place, and a width as 3 cm correct to the nearest cm. Find the upper bound of the area.

- A 16.1 cm^2
- B 11.375 cm^2
- C 16.45 cm^2
- D 16.275 cm^2
- E 14.375 cm^2
- F 13.8 cm^2

3 Expand and simplify $(x + 2)(x - 1)(x + 3)$.

A $x^3 + 4x^2 + 7x - 6$

B $x^3 + 2x^2 - 5x - 6$

C $x^3 - 4x^2 + x + 6$

D $x^3 + 4x^2 - x - 6$

E $x^3 + 6x^2 + 11x + 6$

F $x^3 + 4x^2 + x - 6$

4 The polynomial $x^3 + ax^2 + bx + c$ has roots 2, -1 and 3. Find the value of $a + b + c$.

A 5

B -5

C -3

D 3

E -7

F 7

- 5 A sector of a circle of radius 6 cm has area $15\pi \text{ cm}^2$. Find the arc length of the sector.
- A** $\frac{5\pi}{6} \text{ cm}$
- B** $10\pi \text{ cm}$
- C** $5\pi \text{ cm}$
- D** $\frac{5\pi}{2} \text{ cm}$
- E** $3\pi \text{ cm}$
- F** $15\pi \text{ cm}$

6 Solve $3^{2x} = 4$. Give x in terms of logarithms to base 3.

A $x = \log_3 4 - 2$

B $x = \log_3 2$

C $x = \frac{\log_3 4}{4}$

D $x = \log_2 3$

E $x = \frac{1}{2} \log_3 2$

F $x = 2 \log_3 2$

- 7 The circle with centre $(2, -1)$ passes through $(5, 3)$. Find the equation of the tangent to the circle at $(5, 3)$.

A $4x - 3y = 11$

B $3x + 4y = 27$

C $x + y = 8$

D $3x + 4y = 15$

E $3x - 4y = 3$

F $4x + 3y = 29$

8 Which of the following functions is undefined at exactly two real values of x ?

A $f(x) = \frac{1}{(x-2)^2}$

B $f(x) = \frac{1}{x^2}$

C $f(x) = \sqrt{x-2}$

D $f(x) = \frac{1}{x^2 + 4}$

E $f(x) = \frac{1}{x^2 - 4}$

9 A cuboid measures 3 cm by 4 cm by 12 cm. Find the length of its longest diagonal.

A $\sqrt{153}$ cm

B 13 cm

C 15 cm

D 5 cm

E $\sqrt{119}$ cm

F 12.5 cm

10 Convert 135° to radians.

- A** $\frac{4\pi}{3}$
- B** $\frac{3\pi}{8}$
- C** $\frac{4\pi}{5}$
- D** $\frac{6\pi}{2}$
- E** $\frac{3\pi}{5}$
- F** $\frac{3\pi}{4}$

- 11** A running track consists of two straights of length 100 m and two semicircular ends of radius r . The total length of the track is 400 m. Find r .

A 100π

B $\frac{100}{\pi}$

C $\frac{200}{\pi}$

D $\frac{400}{\pi}$

E $\frac{300}{\pi}$

F $\frac{50}{\pi}$

12 How many real solutions does the equation $x^4 - 5x^2 + 4 = 0$ have?

A 2

B 4

C 5

D 0

E 1

F 3

13 The distance between the parallel lines $3x + 4y = 5$ and $3x + 4y = 20$ is

A 15

B 3

C 4

D $\frac{15}{7}$

E 1

F 5

- 14** Given that $(2 + \sqrt{3})^2 = a + b\sqrt{3}$ and $(2 + \sqrt{3})^{-1} = c + d\sqrt{3}$, where a, b, c, d are integers, find the value of $a + b + c + d$.

A 14

B 4

C 6

D 10

E 12

F 8

- 15** The vertices of a triangle are $A(0, 0)$, $B(6, 0)$ and $C(2, 4)$. Find the equation of the line through C that bisects the area of the triangle.

A $y = x + 2$

B $y = -2x + 8$

C $x = 2$

D $y = 2x$

E $y = 4x - 4$

F $y = -4x + 12$

16 Given that $x = \sqrt{7} + \sqrt{3}$ and $y = \sqrt{7} - \sqrt{3}$, find $x^2 + y^2 - xy$.

A 24

B 16

C 4

D 20

E $20 - 2\sqrt{21}$

F 12

- 17 The volume of a cuboid with a square base of side x is $V = 4x^2 - x^3$ for $0 < x < 4$. Find the maximum volume.

A 16

B $\frac{64}{27}$

C 8

D $\frac{32}{27}$

E $\frac{256}{128}$

F $\frac{27}{27}$

- 18** A ball is dropped from a height of 10 m. After each bounce it rises to $\frac{3}{5}$ of its previous height. Find the total distance travelled by the ball before it comes to rest.
- A** 30 m
- B** 20 m
- C** 40 m
- D** 25 m
- E** 50 m
- F** 35 m

19 Find the value of $(\sqrt{2} + 1)^4 + (\sqrt{2} - 1)^4$.

A 34

B 36

C 17

D $34 + 24\sqrt{2}$

E $24\sqrt{2}$

F 16

- 20** When $p(x)$ is divided by $(x - 1)$ the remainder is 3, and when divided by $(x + 1)$ the remainder is -1 . Find the remainder when $p(x)$ is divided by $(x^2 - 1)$.

- A** 2
- B** $2x + 1$
- C** $-x + 2$
- D** $2x - 1$
- E** $x + 2$
- F** $3x - 1$

END OF TEST